

Memecoin bubbles: resale option or extrapolative demand?

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September 3, 2025

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Keywords: bubbles, Extrapolative model, Resale Option Hypothesis, cryptocurrencies

JEL Classification: C01, G12, G14

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1 Introduction

The analysis of asset price bubbles and investor behavior is not trivial due to identification challenges. Often empirical analysis of bubbles struggle to clearly separate an asset's intrinsic value from its bubble component. The intrinsic value of an asset is often latent and need to be inferred. Therefore, the estimated magnitude of the bubble by construction inherits the deficiencies or biases of the underlying fundamental valuation model used by the researchers. For this reason, Pastor and Veronesi (2006) could argue against the existence of the Nasdaq internet bubble in the late 1990s and Garber (2000) could show rational explanations for the Mississippi and South Sea bubbles. Whilst there are generally accepted valuation models for more traditional asset classes, such as equities, bonds and derivatives, cryptocurrency valuation models are still relatively nascent. Therefore, analyzing cryptocurrency bubbles can be problematic. In this paper, we propose a novel solution, bypassing the fundamental value problem by examining a special classes of cryptocurrencies that, by definition, have negligible fundamental value, *meme-coins*. These classes of cryptocurrencies provide us with a natural experiment where we can examine speculative bubbles with clear identification. We believe that our analysis of memecoins provides reseachers with insights on cryptocurrency investor behavior. In particular, we analyze whether meme coin bubble formation is driven by two key bubble theories: speculation driven *extrapolative demand* (Barberis et al, 2018) and disagreement based *resale option theory* (Scheinkman and Xiong, 2003).

Memecoins cannot be directly used as a medium of exchange, and nor do they provide the holder with any claims to future cashflows or are tied to any fiat currency. Despite this memecoins are traded frequently due to their whimsical appeal and often held by investors that also hold more traditional forms of cryptocurrencies. The largest memecoin by market capitalization, Dogecoin, based on the 'Doge' internet meme, was created in December 2013 as a joke with no use case aside from being a 'fun cryptocurrency', yet by January of 2018 Dogecoin had a 2 billion dollar market capitalization. In this study, we focus on six of the most liquid memecoins that are tradable via large centralized exchanges.

We test whether it is true that the mispricing we observe is largely the value of the resale option investors anticipate due to heterogeneous beliefs and the inability to short (see Harrison and Kreps, 1978 and Scheinkman and Xiong, 2003) or due to speculative extrapolators (see Barberis et al., 2018). This is achieved by examining the empirical relationship between turnover, mispricing and volatility discussed in Cheng, Lung and Wang (2009) and Xiong and Yu (2011). If empirical tests support the presence of a resale option, then we argue that the value ascribed to these memecoins is driven by the value

attributed to the option for the holder to trade them at a later date for Bitcoins, which can subsequently be converted to cash. If there is evidence for extrapolative beliefs, then a significant portion of memecoin trading is reactionary to near term price movements.

The paper is organized as follows. In section two, we discuss the relevant literature associated with resale option theory and extrapolative beliefs. In section three, we lay out the empirical methodology. In section four we discuss the data used and section five presents the results. Section six concludes the paper.

2 Literature Review

Empirical research on cryptocurrency bubbles have largely been focused on Bitcoin (see Kreuser and Sornette, 2018, Li et al., 2019, Huber and Sornette, 2022, Nanez Alonso et al, 2024) rather than memecoins. Furthermore, many papers have largely focused on the role of investor sentiment on cryptocurrency bubbles. For instance, Chowdhury and Damianov (2024) use policy uncertainty indices based on news coverage (developed by Lucey et al., 2022) to explain cryptocurrency bubbles. Li and Yang (2022) analyze the connection between Bitcoin and Ethereum to memecoins. In our study, we focus on two key theoretical models on bubble formation the resale option hypothesis and extrapolative demand. Both models yield testable implications which we examine for memecoins. Our objective is to determine whether memecoin trading is driven by the resale option investor believe that exist or by extrapolative demand from speculators.

2.1 Resale Option Hypothesis

Memecoin pricing can potentially be explained by Miller (1977), Harrison and Kreps (1978) and Scheinkman and Xiong (2003) via heterogeneous expectations. In a market where there is limited or no short-selling, a small group of over-confident or optimistic traders can bid up the price of the asset from intrinsic value. Harrison and Kreps (1978) show that investors can behave in a speculative fashion if they realize they can resell the asset at a later stage to somebody who is willing to pay more than what they would pay if they were to hold it into perpetuity. In Schienkman and Xiong (2003), the option to sell to someone with a higher valuation at a later stage is termed a resale option. Hence, investors of an asset, given a market with short-sell constraints, are no longer willing to pay just the intrinsic value but also the resale option value. The value of this option is positively related to heterogeneity of future beliefs.

Heterogeneity of beliefs and the resale option have been applied in various settings. Hong and Stein (2007) show that the monthly share turnover of Internet stocks had a strong

correlation to their cumulative returns. Cheng, Lung and Wang (2009) show that stock market mispricings are explained better using the resale option hypothesis than the money illusion hypothesis of Modigliani and Cohen (1979). Mei, Schienkman and Xiong (2009) apply the resale option hypothesis to the Chinese A-B share premia, and Xiong and Yu (2011) apply it to mispriced out of the money Chinese warrants. In this paper, we apply it to memecoins.

As discussed in Chen, Lung and Wang (2009), the resale option hypothesis yields two testable implications. First, the level of the mispricing should be positively related to the heterogeneity of beliefs among investors, which is manifested through trading turnover. Second, the volatility of the mispricing should also increase with the degree of heterogeneous beliefs. We test both these relationships for memecoins.

2.2 Extrapolative Beliefs

Barberis et al. (2018) show that bubble formation is driven by the extrapolative beliefs of speculators. In their setting, the market is dominated by extrapolators with varying degrees of “wavering” and fundamental traders are not present as a correcting force. This potentially might be more relevant for memecoin traders as it can be argued that fundamental value does not exist for these assets. The heterogeneity in wavering provides the ability for extrapolators to trade among themselves and create high trading volume. This differs from the heterogeneous belief in valuation in the Scheinkman and Xiong (2003) setting.

An extrapolator buys assets whose prices have risen because they expect the prices to continue rising. Due to this, there is a positive relationship between past returns and volume. According to Barberis et al. (2018), with Scheinkman and Xiong (2003), an increase in disagreement among investors on valuation leads to an contemporaneous increase in prices, volume (turnover) and volatility, and no lagged effect. Examining whether there is a lagged effect between returns and volume will be central to our empirical analysis.

3 Methodology

We perform tests to identify whether extrapolative demand or resale option demand dominate memecoin markets. At the heart of this, is the to identify whether prices and volume spikes in memecoins comes simultaneously or whether there is some lead lag dynamic. According to Barberis et al. (2018), we should expect high trading volume to lag spikes in prices, as the varying degrees of ‘wavering’ between extrapolators generate a spike in volume. In Scheinkman and Xiong (2003), it is the valuation disagreement that leads to both higher volumes and overpricing. The relationship between volumes

and prices are contemporaneous.¹

We begin with the formulation suggested by Xiong and Yu (2010) and Chen, Lung and Wang (2009) on testing the Scheinkman and Xiong (2003) resale option hypothesis. The resale option hypothesis implies that both the level of mispricing ($\epsilon_{i,t}$) is positively related to the heterogenous beliefs, which is proxied using cryptocurrency turnover. Turnover is defined as,

$$\text{TURNOVER}_{i,t} = \frac{\text{VOLUME}_{i,t} \times \text{PRICE}_{i,t}}{\text{MARKET CAP}_{i,t}}$$

When there is no significant issuance of new cryptocurrency, this is not substantially different to volume as described in Barberis et al. (2018).

$$\epsilon_{i,t} = b_0 + b_1 \text{TURNOVER}_{i,t} + u_{i,t} \quad (1)$$

According to Chen, Lung and Wang (2009) the resale option hypothesis implies that $b_1 > 0$. However, this approach can be problematic for cryptocurrencies as mispricing $\epsilon_{i,t}$ is often non-stationary. In the context of memecoins, the mispricing is simply the log price. $\epsilon_{i,t} = \ln P_{i,t} = \sum_{k=0}^t \ln R_{i,k}$ where $R_{i,t}$ is the gross return. We can substitute this into equation 1 as well as flip the regression so that turnover is now the dependent variable.

$$\text{TURNOVER}_{i,t} = c_0 + c_1 \left(\sum_{k=0}^t \ln R_{i,k} \right) + v_{i,t} \quad (2)$$

where $c_1 = b_1 \frac{\sigma_p^2}{\sigma_\epsilon^2}$. We can generalize to the following reduced-form, where the coefficients to the lagged returns are allowed to vary. $\text{TURNOVER}_{i,t} = c_0 + c_1 \ln R_{i,t} + c_2 \ln R_{i,t-1} + c_3 \ln R_{i,t-2} + \dots + v_{i,t}$. Therefore instead of testing for the relationship between prices and turnover, as suggested in Chen, Lung and Wang (2009), we can test between returns and turnover. This bypasses the non-stationary problem with prices.

In our empirical tests, we examine up to 5 lags and form a panel regression with cryptocurrency fixed effects to account for potential differences in turnover between various memecoins.

$$\text{TURNOVER}_{i,t} = \sum_{k=0}^5 c_k \ln R_{i,t-k} + \text{FIXED EFFECT}_i + v_{i,t} \quad (3)$$

The Barberis et al. (2018) extrapolation model predicts a strong positive correlation between volume and past returns during bubble episodes. This can be tested with equation 3. Barberis et al. (2018) also show that under the Scheinkman and Xiong (2003) resale option hypothesis, only c_0 would be positive, and lagged returns would be uncorrelated to

¹In contrast, traditional rational bubble models (Tirole, 1985) do not explain the relationship between prices and volumes.

volume. Therefore, by testing whether $c_1, c_2, \dots, c_5$ are statistically significant allows us to determine whether memecoin traders are speculative with extrapolative expectations (as modelled in Barberis et al., 2018). Furthermore, we also test the same relationship for Bitcoin. This allows us to compare and contrast the trading dynamics of memecoins relative to Bitcoin.

In extension to examining the relationship between turnover and past returns, we also examine the relationship between turnover and lagged volatility. Returns are directional, whilst volatility is not. In equation 3, negative returns would imply lower turnover, as extrapolative speculators are less likely to trade following negative returns in price. In Barberis et al. (2018) extrapolators were largely concerned with positive returns and how speculators with varying degrees of confidence interacted through trading. Here, we entertain the idea that speculators in memecoins might also be compelled to trade following increased volatility. In Schienkman and Xiong (2003) and Xiong and Yu (2011) mispricing volatility increases when there is greater heterogeneity in beliefs and that the bubble is positively correlated to its volatility. In this paper, we empirically test whether past volatility positively impacts turnover using a fixed effects panel regression.

$$\text{TURNOVER}_{i,t} = \sum_{k=0}^5 c_k RV_{i,t-k} + \text{FIXED EFFECT}_i + v_{i,t} \quad (4)$$

where $RV_{i,t} = \sqrt{\sum_{j=1}^J r_{i,j}^2}$ is the sum of squared returns in 30 minute trading intervals across day t for memecoin i obtained from Binance. We also compare and contrast our memecoin results with Bitcoin to examine if there is a significant difference between memecoin trading behavior and more mainstream cryptocurrencies. We expect to see memecoin trading to exhibit more of the characteristics discussed in Scheinkman and Xiong (2003) and Barberis et al. (2018).

4 Data

We examine six of the largest memecoins that are trade on the largest centralized exchange, Binance from January 2020 to September 2024. We examine memecoin-USDT pairs. The six memecoins are: Dogecoin, Shiba Inu, Floki, Pepe, Memecoin and Bonk. For daily market capitalization data, we use CoinMarketCap. For price volume data, we use aggregate OHLC 30 minute trading bins downloaded from Binance. Memecoin realized volatility is calculated as the sum of squared returns over these 30 minute intervals across the day. In Table 1, we provide the summary statistic for our six memecoins, as well as Bitcoin (as a benchmark comparison). We examine the distributional characteristics of their daily returns, turnover and realized volatility.

[Insert Table 1]

5 Empirical Results

Scheinkman and Xiong (2003) and Hong, Scheinkman and Xiong (2006) show that excess optimism causes greater turnover and a rise in the resale option. Table 2 (column 1) documents a positive contemporaneous relationship between turnover and returns from our fixed effects panel regression with cluster errors. However, we note that lagged returns are statistically significant and positive. This characteristic is not consistent with Scheinkman and Xiong (2003) but strongly predicts Barberis et al. (2018) where past returns are positively correlated with future turnover. As such, we believe memecoin traders are likely to be speculators that have extrapolative demand based on past price changes. This is only true for our memecoins, and we did not find any significance in lagged or contemporaneous returns for Bitcoin (column 2). Therefore, Bitcoin trading does not behave with bubble-like characteristics.

In extension, we also added lagged turnover to our regression specification (column 3 and 4). We show that the positive relationship between lagged returns and turnover remains robust for memecoins, even after accounting for lagged turnover.

[Insert Table 2]

In Table 3, we examine the impact of lagged realized volatility on turnover for memecoins. We show that realized volatility is a strong predictor of turnover. This suggests that memecoin trading increases during periods of heightened volatility. Whilst lagged coefficients also show significance, this disappears when we incorporate lagged turnover in our panel regression. Overall, we find that both Bitcoin and memecoins exhibit a positive relationship, suggesting that heterogeneity of beliefs among investors (and greater trading) does occur during periods of high volatility. Therefore, the resale option hypothesis seems to be present there in both memecoins and mainstream cryptocurrencies such as Bitcoin. The lack of significance in lagged volatility is intuitive. Whilst cryptocurrency traders are attracted to volatility, they are not extrapolating based on past volatility. Extrapolative demand only is evident with regards to past returns, and on this point, it impacts only memecoins and not Bitcoin.

[Insert Table 3]

6 Conclusion

Both the resale option hypothesis (Scheinkman and Xiong, 2003) and extrapolative demand model (Barberis et al., 2018) on asset pricing bubbles yield testable empirical predictions on the relationship between turnover and past returns. We examine six of the most liquid memecoins and show that lagged returns has a positive and statistically significant relationship with turnover. We also find that this is not true for Bitcoin, suggesting that speculators with extrapolative demand that drive memecoin pricing are less present in Bitcoin trading. This also suggests that memecoin trading dynamics are indeed different to Bitcoin and more prone to bubble-like properties. Whilst our empirical results favor Barberis et al. (2018) in explaining memecoins, we also note that both Bitcoin and memecoins show a positive relationship between volatility and turnover suggesting that heterogeneity of beliefs among cryptocurrency investors does increase during periods of heightened volatility.

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Table 1: **Summary Statistics**

This table presents the descriptive statistics of the memecoin we analyse, along with the Bitcoin benchmark. We document the high, low, average market capitalizations of our sample in USD in Panel A. We document summary statistics for memecoin daily returns, turnover and realized volatility in Panel B. All data was sourced from the centralized exchange Binance.

<i>Panel A</i>						
symbol	Start_Date	End_Date	Obs	Market Capitalization		
				Mean	Min	Max
BONKUSDT	2023-12-15	2024-09-11	272	1,410,650,035	605,305,815	2,726,057,155
DOGEUSDT	2020-01-01	2024-09-05	1,708	13,499,454,008	183,588,292	88,794,768,977
FIOKIUSDT	2023-05-06	2024-09-17	501	862,862,262	153,609,915	3,121,649,595
MEMEUSDT	2023-11-03	2024-09-11	314	349,471,242	125,391,098	757,413,426
PEPEUSDT	2023-05-06	2024-09-17	501	1,862,702,319	257,277,784	6,994,599,136
SHIBUSDT	2021-05-16	2024-09-21	1,225	8,688,098,801	2,947,395,870	41,191,243,533
BTCUSDT	2020-01-01	2024-09-05	1,708	640,300,523,858	91,962,797,384	1,436,630,000,000

<i>Panel B</i>								
Symbol	Variable	Mean	SD	25th perc.	Median	75th perc.	Min	Max
BONKUSDT	Returns	-0.002	0.082	-0.048	0	0.035	-0.191	0.336
	Turnover	0.055	0.047	0.025	0.040	0.068	0.007	0.292
	Realized Volatility	0.077	0.042	0.050	0.066	0.090	0.025	0.355
DOGEUSDT	Returns	0.002	0.076	-0.022	-0.0004	0.019	-0.495	1.506
	Turnover	0.0004	0.001	0.0001	0.0001	0.0003	0.00000	0.024
	Realized Volatility	0.037	0.066	0.009	0.021	0.042	0	1.506
FIOKIUSDT	Returns	0.002	0.072	-0.030	-0.003	0.021	-0.258	0.759
	Turnover	0.040	0.057	0.011	0.022	0.043	0.003	0.628
	Realized Volatility	0.062	0.041	0.035	0.051	0.073	0.017	0.297
MEMEUSDT	Returns	-0.003	0.069	-0.036	-0.004	0.029	-0.205	0.378
	Turnover	0.138	0.233	0.023	0.048	0.138	0.004	1.645
	Realized Volatility	0.067	0.040	0.043	0.055	0.077	0.019	0.270
PEPEUSDT	Returns	0.002	0.080	-0.036	0	0.031	-0.279	0.424
	Turnover	0.084	0.089	0.034	0.057	0.100	0.009	0.743
	Realized Volatility	0.082	0.041	0.059	0.072	0.089	0.016	0.286
SHIBUSDT	Returns	-0.0001	0.064	-0.025	-0.003	0.019	-0.536	0.511
	Turnover	0.018	0.038	0.005	0.008	0.017	0.001	0.598
	Realized Volatility	0.055	0.041	0.032	0.044	0.063	0.010	0.402
BTCUSDT	Returns	0.001	0.034	-0.012	0.0004	0.016	-0.273	0.178
	Turnover	0.0001	0.0002	0.00004	0.0001	0.0001	0.00001	0.002
	Realized Volatility	0.023	0.026	0.006	0.015	0.031	0	0.273

Table 2: The Effect of Lagged Returns on Turnover

In (1) we present a fixed effects panel regression with clustered errors for memecoins to test whether lagged returns has an impact on turnover. We find that both contemporaneous returns and lagged returns have a positive relationship with turnover. In comparison, we do not see this effect for Bitcoin (2). In (3) we account for lagged turnover and show that lagged returns still explains memecoin turnover. This suggests that Barberis et al. (2018) extrapolative demand exists for memecoins but is not detected with Bitcoin. In (4) we show that whilst Bitcoin turnover is autoregressive, it is not explained by lagged returns. Therefore, while memecoins exhibit bubble-like properties described in Barberis et al. (2018), Bitcoin does not.

	<i>Dependent variable:</i>			
	Turnover			
	<i>Memecoins</i>	<i>Bitcoin</i>	<i>Memecoins</i>	<i>Bitcoin</i>
	(1)	(2)	(3)	(4)
Turnover Lag(1)			0.622*** (0.015)	0.417*** (0.024)
Turnover Lag(2)			0.012 (0.017)	0.071*** (0.026)
Turnover Lag(3)			0.112*** (0.017)	0.123*** (0.026)
Turnover Lag(4)			0.031** (0.016)	0.093*** (0.026)
Turnover Lag(5)			0.077*** (0.013)	0.105*** (0.024)
Returns	0.233*** (0.013)	0.0001 (0.0001)	0.209*** (0.008)	0.0001 (0.0001)
Returns Lag(1)	0.175*** (0.013)	0.0001 (0.0001)	0.027*** (0.009)	0.0001 (0.0001)
Returns Lag(2)	0.116*** (0.013)	-0.0001 (0.0001)	0.006 (0.008)	-0.0002** (0.0001)
Returns Lag(3)	0.087*** (0.013)	0.00003 (0.0001)	-0.013 (0.008)	0.0001 (0.0001)
Returns Lag(4)	0.068*** (0.013)	-0.0001 (0.0001)	-0.018** (0.008)	-0.0002** (0.0001)
Returns Lag(5)	0.062*** (0.013)	-0.0002* (0.0001)	-0.018** (0.008)	-0.0001* (0.0001)
Constant		0.0001*** (0.00000)		0.00002*** (0.00000)
Cryptocurrency FE	Yes	N/A	Yes	N/A
Observations	4,491	1,703	4,491	1,703
R ²	0.137	0.005	0.691	0.450
Adjusted R ²	0.135	0.002	0.690	0.446
F Statistic	118.376***	1.502	910.656***	125.713***

Note:

*p<0.1; **p<0.05; ***p<0.01

Table 3: **The Effect of Lagged Realized Volatility on Turnover**

In (1) we present a fixed effects panel regression with clustered errors for memecoins to test whether lagged realized volatility has an impact on turnover. We find that both contemporaneous returns and lagged realized volatility have a positive relationship with turnover. In (2) we run a regression for Bitcoin, and find similar results. In (3) and (4) we account for lagged turnover and show that most of the lagged volatility significance is reduced.

	<i>Dependent variable:</i>			
	Turnover			
	<i>Memecoins</i>	<i>Bitcoin</i>	<i>Memecoins</i>	<i>Bitcoin</i>
	(1)	(2)	(3)	(4)
Turnover Lag(1)			0.618*** (0.015)	0.382*** (0.024)
Turnover Lag(1)			0.031* (0.017)	0.085*** (0.026)
Turnover Lag(1)			0.079*** (0.017)	0.135*** (0.026)
Turnover Lag(1)			0.027* (0.016)	0.093*** (0.026)
Turnover Lag(1)			0.100*** (0.014)	0.128*** (0.024)
Volatility	0.411*** (0.022)	0.001*** (0.0001)	0.308*** (0.014)	0.001*** (0.0001)
Volatility Lag(1)	0.108*** (0.023)	0.0004*** (0.0001)	-0.125*** (0.015)	-0.0001 (0.0001)
Volatility Lag(2)	0.057** (0.023)	0.0002* (0.0001)	-0.032** (0.015)	-0.0001 (0.0001)
Volatility Lag(3)	0.056** (0.023)	-0.00001 (0.0001)	-0.009 (0.015)	-0.0003*** (0.0001)
Volatility Lag(4)	0.052** (0.023)	0.0001 (0.0001)	-0.019 (0.015)	-0.0002 (0.0001)
Volatility Lag(5)	0.049** (0.021)	-0.0001 (0.0001)	-0.049*** (0.014)	-0.0004*** (0.0001)
Constant		0.0001*** (0.00001)		0.00002*** (0.00001)
Cryptocurrency FE	Yes	N/A	Yes	N/A
Observations	4,491	1,703	4,491	1,703
R ²	0.175	0.077	0.680	0.496
Adjusted R ²	0.173	0.073	0.679	0.492
F Statistic	158.744***	23.463***	866.084***	151.111***
<i>Note:</i>		*p<0.1; **p<0.05; ***p<0.01		